CSCI 131 — Introduction to Programming in Python
Programming Assignment 11
Animation: Ball and Paddle

On Wednesday we made a ball bounce off the walls (`walls.py`) and a paddle that moves with the arrow keys (`paddle.py`). Today you turn them into a one-player game: keep the ball in the air with the paddle for as long as you can.

Download the starter file `paddle_game.py` from the **Lab 11** page on the course site, and save it in the **same folder** as `anim.py`. Open it in IDLE and put your name on the first line.

The starter is the code we wrote on Wednesday: the ball bounces around all four walls, and the paddle moves with the left and right arrow keys. Read through the starter and find the part that moves the paddle, the part that moves the ball, and the four `if`s for the walls.

Do the three parts in order, and run your program after each one.

Part 1 — Through the floor

The floor should no longer be a wall. Change the program so that the ball falls out through the bottom of the window instead of bouncing off it.

When the ball has gone **completely** out of the bottom of the window, the game is over: the loop must stop, so the program reaches `win.done()` and the window stays open showing where everything was. You can end a `while` loop with `break`, or by adding a second condition to the `while` line with `and`.

Remember that the ball is placed by its **centre**: its top edge is `y - size` and its bottom edge is `y + size`. Think about which edge has to be past `win.height` for the ball to be completely out of the window.

Part 2 — Bounce off the paddle

Make the ball bounce off the paddle. The ball **hits** the paddle when both of these are true:

- the bottom of the ball is below the top of the paddle;
- the ball's `x` is between the paddle's left edge and its right edge.

The paddle is placed by its **top-left corner**, so its top edge is `py`, its left edge is `px` and its right edge is `px + pw`.

When the ball hits the paddle, do what the walls do: put the ball exactly on top of the paddle, then flip the sign of `dy` so it goes back up.

Part 3 — Keep score

Add a score that goes up by one every time the ball bounces off the paddle. The score starts at `0` and is shown in the info panel in the top left. Use `win.set_text` to change it.

Submission

Submit `paddle_game.py` on Canvas by the end of the period. Put your name in the comment on the first line. If a classmate helped you, put their name in the comment as well.